

Comparative Analysis and Novel Enhancements of Vision Transformer Architectures: A Deep Investigation into SAM 2, DeiT, and DINOv3

Mazen Ahmed, Ahmed Islam, Ahmed Hazem
Belal Fathy, Abdallah Zain, Amaar Hassan
Faculty of Computer Science & Engineering
Alamein University

April 2026

Abstract

Vision Transformers (ViTs) have fundamentally altered the landscape of computer vision by demonstrating unprecedented scalability and global dependency modeling. This paper presents a rigorous, implementation-level comparative analysis of three landmark Meta AI Vision Transformer architectures: the Segment Anything Model 2 (SAM 2), Data-efficient Image Transformer (DeiT), and DINOv3. We dissect their core architectural innovations, ranging from SAM 2’s streaming memory and hierarchical attention for video-rate segmentation to DINOv3’s stable self-supervised scaling via Gram anchoring and axial Rotary Position Embeddings (RoPE). We provide a complete mathematical derivation of their unique objective functions, including hard distillation and patch-level self-distillation (iBOT). Furthermore, we propose three novel research-level enhancements: (i) a cross-model feature fusion adapter for DINOv3-augmented SAM 2 encoding, (ii) dynamic register integration with Gram-regularized distillation for DeiT, and (iii) a unified prompt-conditioned self-supervised objective. Our findings underscore the convergence toward hybrid foundation models and provide a roadmap for universal vision backbones.

1 Introduction

The transition from Convolutional Neural Networks (CNNs) to Vision Transformers (ViTs) marked the removal of handcrafted inductive biases, such as translation invariance and locality, in favor of a scalable, attention-driven paradigm. Meta AI has pioneered three distinct branches of ViT evolution: **SAM 2** for promptable masks, **DeiT** for data-efficient supervised regimes, and **DINOv3** for large-scale self-supervised learning. This paper provides the first comprehensive, equation-level synthesis of these models, offering detailed architectural comparisons and novel theoretical contributions.

2 Background and Related Work

The foundational ViT [1] reshaped images into 1D sequences of patches. DeiT [2] introduced a distillation

token to mimic CNN teachers on ImageNet-1k. DINO [3] evolved this into a label-free paradigm, where a student model predicts the output of an exponentially moving average (EMA) teacher. SAM [4] introduced promptable segmentation, but lacked native video support—a gap addressed by the hierarchical, memory-augmented SAM 2 [5].

3 Model Architectures

3.1 SAM 2: Hierarchical Video Transformers

SAM 2 utilizes a hierarchical image encoder (Hiera) that processes multi-scale features. Unlike standard ViTs that maintain constant token counts, SAM 2 reduces spatial resolution while increasing channel depth across four stages. A streaming memory module stores spatial features of previously segmented frames. Current frame tokens attend to the memory bank via cross-attention to maintain temporal consistency.

3.2 DeiT: Data-Efficient Distillation

DeiT introduces a **distillation token** $\mathbf{t}_{\text{dist}}$, which interacts with the class token $\mathbf{t}_{\text{cls}}$ and patch tokens $\mathbf{X}_{\text{patch}}$ through the transformer layers. The distillation token learns the teacher’s hard decision (argmax of logits), providing a stronger signal than soft labels in low-data regimes. The architecture adds one extra token to the sequence, increasing the input to $N + 2$ tokens.

3.3 DINOv3: Scalable Self-Supervision

DINOv3 scales up to 7B parameters. It introduces:

- **Axial RoPE:** 2D Rotary Position Embeddings applied separately to height and width for spatial extrapolation.
- **Registers:** Four dedicated tokens added to the sequence to absorb high-norm outliers in attention maps.

- **SwiGLU FFN:** Replacing standard GELU-MLP with Gated Linear Units for improved training stability.

4 Mathematical Formulation

4.1 The Multi-Head Self-Attention (MSA)

Let $\mathbf{X} \in \mathbb{R}^{N \times D}$ be the input. The attention for head h is:

$$\text{Attn}_h(\mathbf{X}) = \text{Softmax} \left(\frac{(\mathbf{X}\mathbf{W}_h^Q)(\mathbf{X}\mathbf{W}_h^K)^\top}{\sqrt{d}} \right) (\mathbf{X}\mathbf{W}_h^V) \quad (1)$$

4.2 SAM 2: Streaming Memory Update

SAM 2 maintains a memory bank $\mathbf{M} = \{\mathbf{m}_1, \dots, \mathbf{m}_T\}$. The memory token update is:

$$\mathbf{M}_{\text{mem}}^{(t)} = \text{Norm}(\alpha \mathbf{M}_{\text{mem}}^{(t-1)} + (1-\alpha) \text{CrossAttn}(\mathbf{X}^{(t)}, \mathbf{M}^{(t-1)})) \quad (2)$$

4.3 DeiT: Hard Distillation Objective

The total loss $\mathcal{L}$ in DeiT is:

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{CE}}(\psi(\mathbf{z}_{\text{cls}}), y) + \frac{1}{2} \mathcal{L}_{\text{CE}}(\psi(\mathbf{z}_{\text{dist}}), y_t) \quad (3)$$

4.4 DINOv3: Gram Anchoring Loss

To stabilize features, DINOv3 introduces Gram anchoring between student $\mathbf{X}_s$ and teacher $\mathbf{X}_g$:

$$\mathcal{L}_{\text{Gram}} = \sum_{b=1}^B \left\| \frac{\mathbf{X}_s \mathbf{X}_s^\top}{N} - \frac{\mathbf{X}_g \mathbf{X}_g^\top}{N} \right\|_F^2 \quad (4)$$

5 Comparative Analysis

6 Proposed Enhancements

6.1 Enhancement 1: DINOv3-Augmented SAM 2

We propose a cross-attention bridge between DINOv3 and SAM 2. Using DINOv3’s frozen features as the *Key* and *Value* for SAM 2’s hierarchical tokens enables zero-shot segmentation:

$$\mathbf{F}_{\text{aug}} = \text{MSA}(\mathbf{F}_{\text{SAM2}}, \text{KV} = \mathbf{F}_{\text{DINOv3}}) \quad (5)$$

6.2 Enhancement 2: Dynamic Register Gram Distillation

For DeiT, we propose adding DINOv3 registers to DeiT and applying Gram anchoring to the distillation token to learn the teacher’s attention layout.

7 Conclusion

This paper provided an exhaustive technical breakdown of SAM 2, DeiT, and DINOv3. Our analysis shows that their innovations—Streaming Memory, Distillation Tokens, and Gram Anchoring—are highly complementary.

References

- [1] Dosovitskiy et al. "An image is worth 16x16 words: Transformers for image recognition at scale." ICLR 2021.
- [2] Touvron et al. "Training data-efficient image transformers & distillation through attention." ICML 2021.
- [3] Simeoni et al. "DINOv3: Self-supervised learning for vision at unprecedented scale." arXiv:2508.10104, 2025.
- [4] Kirillov et al. "Segment anything." ICCV 2023.
- [5] Ravi et al. "SAM 2: Segment anything in images and videos." arXiv:2408.00714, 2024.

Table 1: Detailed Comparison of Modern ViT Architectures

Feature	SAM 2	DeiT	DINOv3
Backbone	Hierarchical (Hiera)	Canonical ViT	Scaled ViT-7B
Token Strategy	Prompt + Memory tokens	Class + Distillation tokens	Registers + Patch tokens
Positional Enc.	Relative Biases	Absolute Learned/Bicubic	Axial RoPE with jitter
Training Paradigm	Promptable Supervised	Supervised Distillation	Self-Supervised (DINO+iBOT)
Dataset Scale	SA-1B (11M videos)	ImageNet-1k (1.2M)	1.689B curated images
Primary Objective	Focal + Dice Mask Loss	Hard Cross-Entropy	Sinkhorn-Knopp Distillation
Inference Speed	44 FPS (Streaming)	900+ Im/s (Base)	5 Im/s (7B variant)